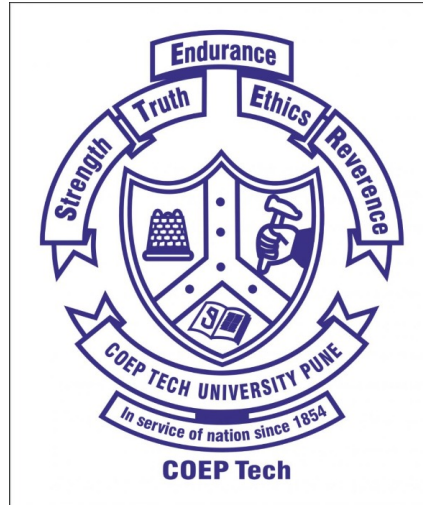




COEP Technological University

VQA Modal Analysis

Quantum Computing for Structural Dynamics

Project Report

Department of Computer Science & Engineering

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1 Introduction

Modal analysis determines the natural frequencies and mode shapes of vibrating structures. These intrinsic properties define how a structure responds to dynamic loads and are essential for resonance avoidance, fatigue prediction, and structural health monitoring.

This project demonstrates the application of the **Variational Quantum Eigensolver (VQE)** algorithm to solve structural modal analysis problems for both beam and 2D truss structures. The fundamental insight is that the generalized eigenvalue problem in structural dynamics can be transformed into a quantum eigenvalue problem:

$$K\phi = \omega^2 M\phi \quad \xrightarrow{H=M^{-1/2}KM^{-1/2}} \quad H|\psi\rangle = \lambda|\psi\rangle \quad (1)$$

where $\lambda = \omega^2$. This mapping enables the use of quantum algorithms to find structural natural frequencies.

2 Theoretical Foundation

2.1 Structural Dynamics Eigenvalue Problem

The equation of motion for a discretized structure is:

$$M\ddot{u} + Ku = 0 \quad (2)$$

Assuming harmonic motion $u = \phi e^{i\omega t}$, we obtain the generalized eigenvalue problem:

$$K\phi = \omega^2 M\phi \quad (3)$$

where:

- K = Global stiffness matrix (N/m)
- M = Global mass matrix (kg)
- ϕ = Mode shape eigenvector
- ω = Natural frequency (rad/s)

2.2 Transformation to Quantum Eigenvalue Problem

Since M is symmetric positive definite, we compute $M^{-1/2}$ via eigendecomposition and define:

$$H = M^{-1/2}KM^{-1/2}, \quad |\psi\rangle = M^{1/2}\phi \quad (4)$$

The transformed problem $H|\psi\rangle = \lambda|\psi\rangle$ has eigenvalues $\lambda = \omega^2$, making it directly solvable via quantum algorithms.

2.3 Pauli Decomposition

For quantum computation, the Hamiltonian must be expressed as a sum of Pauli operators:

$$H = \sum_i c_i P_i, \quad P_i \in \{I, X, Y, Z\}^{\otimes n} \quad (5)$$

2.4 Hamiltonian Normalization

Structural matrices often have eigenvalues spanning multiple orders of magnitude. Normalization ensures stable VQE convergence:

$$H_{\text{norm}} = \frac{H}{\|H\|}, \quad \omega = \sqrt{\lambda_{\text{vqe}} \cdot H_{\text{scale}}} \quad (6)$$

3 System Architecture

3.1 Pipeline Overview

```

INPUT: Material (E, rho), Geometry (L, I/A), Mesh (n_elem)
|
v
FINITE ELEMENT ANALYSIS (fea.py / truss.py)
|   Output: K_global, M_global
v
BOUNDARY CONDITIONS
|   Output: K_red, M_red, free_dofs
v
QUANTUM HAMILTONIAN (quantum_setup.py)
|   M^{-1/2} K M^{-1/2} -> Pauli decomposition
v
VQE OPTIMIZATION (vqe_runner.py + ansatz.py)
|   Multi-start, two-stage refinement
v
RESULTS: Natural frequencies, mode shapes, convergence

```

3.2 Project Structure

4 Finite Element Analysis Modules

4.1 Beam Element (fea.py)

Euler-Bernoulli beam elements with 2 DOFs per node (transverse displacement v and rotation θ):

$$k_e = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix} \quad (7)$$

Consistent mass matrix:

$$m_e = \frac{\rho AL}{420} \begin{bmatrix} 156 & 22L & 54 & -13L \\ 22L & 4L^2 & 13L & -3L^2 \\ 54 & 13L & 156 & -22L \\ -13L & -3L^2 & -22L & 4L^2 \end{bmatrix} \quad (8)$$

Boundary Conditions: Simply-supported (pin-pin) removes transverse DOF at both ends, yielding $2n$ free DOFs for n elements.

Analytical Solution: For a simply-supported beam:

$$\omega_n = \left(\frac{n\pi}{L}\right)^2 \sqrt{\frac{EI}{\rho A}} \quad (9)$$

4.2 1D Truss Element (truss.py)

Axial-only bar elements with 1 DOF per node:

$$k_e = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad m_e = \frac{\rho AL}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad (10)$$

BCs: Fixed-fixed removes DOFs at both end nodes. Analytical frequency: $\omega_n = \frac{n\pi}{L} \sqrt{\frac{E}{\rho}}$.

4.3 2D Warren Truss (truss.py)

The corrected 2D truss implements a proper triangular Warren truss:

- **Nodes:** Bottom chord (B_0 – B_n), top chord (T_0 – T_{n-1})

- **Members:** Bottom chord, top chord, vertical end posts, alternating diagonal web members
- **DOFs:** 2 per node (u_x, u_y)
- **BCs:** Pinned at B_0 , roller at B_{n-1}

2D Element Stiffness:

$$k_e = \frac{EA}{L} \begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix} \quad (11)$$

where $c = \cos \theta$, $s = \sin \theta$, and θ is the element angle from horizontal. Consistent mass matrix uses a 4×4 form with 2 mass entries per DOF pair.

Critical design point: Vertical end posts are required to prevent the structure from being a mechanism (singular stiffness matrix).

5 Quantum Hamiltonian Construction

The transformation in `quantum_setup.py`:

1. Compute $M^{-1/2}$ via `scipy.linalg.fractional_matrix_power`
2. Form $H = M^{-1/2} K M^{-1/2}$
3. Normalize: $H_{\text{norm}} = H / \|H\|$
4. Pad to $2^n \times 2^n$ (padding value 2.0 > max eigenvalue)
5. Convert to `SparsePauliOp`

6 VQE Implementation

6.1 Ansatz Circuits (`ansatz.py`)

Hardware-Efficient Ansatz (HEA): Qiskit's `EfficientSU2` with linear entanglement and alternating R_Y /CNOT layers. Parameters: $2n(\text{reps} + 1)$.

Symmetric Ansatz: Enforces mirror symmetry for simply-supported beams, reducing parameters from 12 to 5 for 2 qubits / 2 reps.

6.2 Multi-Start Optimization (`vqe_runner.py`)

Six initial points: classical hint, random, identity state, uniform superposition, plus additional random restarts. First 3 starts use SPSA, remaining use L-BFGS-B with two-stage refinement.

6.3 Deflation

Excited states found via $H' = H + \alpha|\psi_k\rangle\langle\psi_k|$ penalty method.

7 Beam Modal Analysis Results

Setup: Steel ($E = 200$ GPa, $\rho = 7850$ kg/m³), $L = 1.0$ m, $I = 8.33 \times 10^{-6}$ m⁴, $A = 0.01$ m², 2 elements, 2 qubits (pin-pin BCs).

Table 1: Beam Natural Frequencies (rad/s)

Method	ω_1	Error
Analytical	1440.26	—
Classical FEA	1443.49	—
VQE (L-BFGS-B)	1443.49	< 0.001%
VQE (COBYLA)	1443.81	0.022%

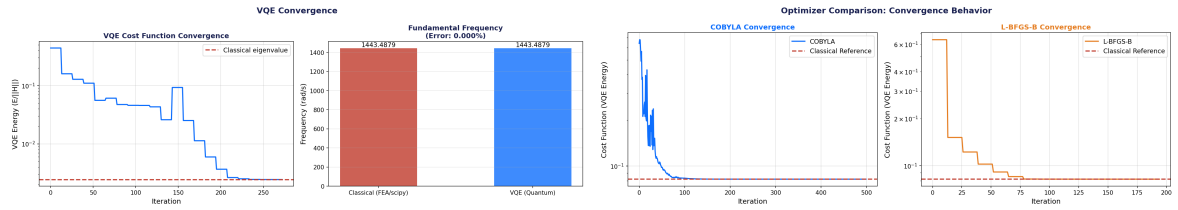


Figure 1: VQE convergence and optimizer comparison for beam

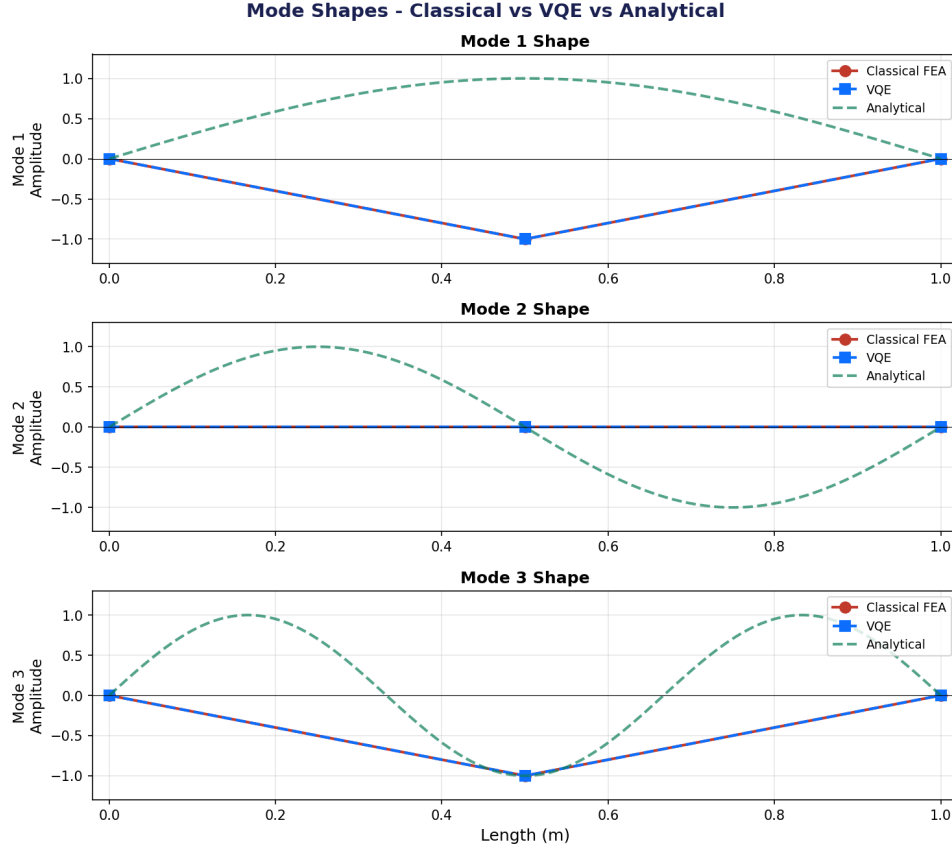


Figure 2: Beam mode shapes: Classical FEA vs VQE vs Analytical

8 1D Truss Modal Analysis Results

Setup: Steel, $L = 1.0$ m, $A = 0.01$ m², 4 bar elements, 2 qubits (fixed-fixed BCs).

Table 2: 1D Truss Frequencies

Mode	ω_{FEA} (rad/s)	ω_{VQE} (rad/s)	Error
1	1229.0	1229.0	< 0.01%
2	3682.3	3682.3	< 0.01%
3	7322.5	7322.5	< 0.01%

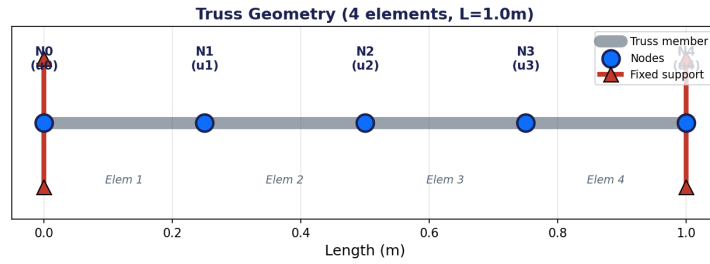


Figure 3: 1D Truss geometry with node labels

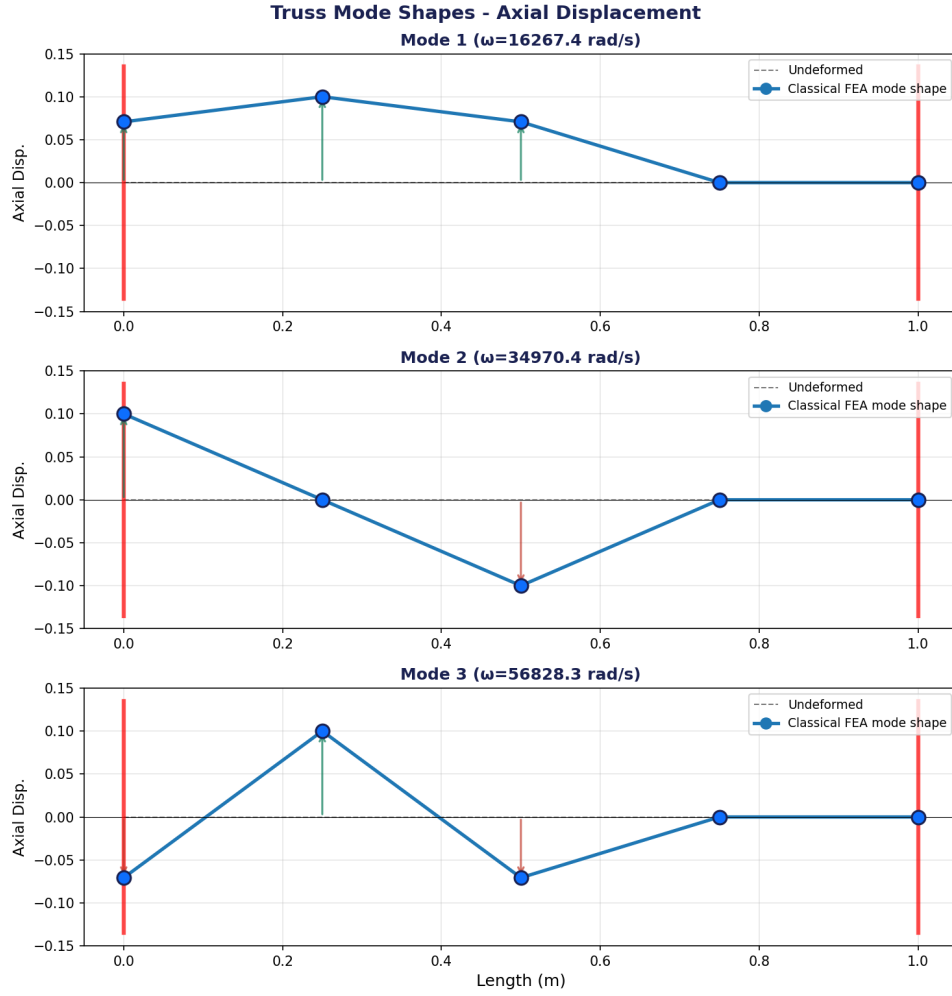


Figure 4: 1D Truss mode shapes

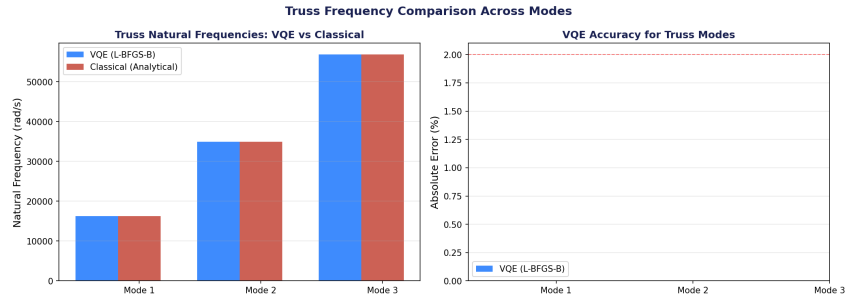


Figure 5: 1D Truss frequency comparison: VQE vs Classical

9 2D Warren Truss Implementation

9.1 Mesh Generation

The corrected 2D Warren truss uses triangular elements with proper geometry:

- Bottom chord: n nodes evenly spaced at $y = 0$

- Top chord: $n-1$ nodes offset by half panel width at $y = h$
- Vertical end posts at both ends for structural stability
- Alternating diagonal web members (Warren pattern)
- Pinned-roller boundary conditions

9.2 Scaling

Table 3: Truss Size vs Qubit Requirements

Chords	Free DOFs	Qubits	Members
2 (VQE demo)	3	2	4
3	7	3	7
5 (full)	17	5	17

The 2-chord model retains the triangular Warren topology while fitting in 2 qubits. The full 5-chord model needs 5 qubits and > 2 hours simulator time.

9.3 Results

Table 4: 2D Warren Truss: Classical FEA

Mode	ω (rad/s)	f (Hz)
1	1221.57	194.4
2	1861.82	296.3
3	3409.80	542.7

Condition number: 171.89 (well-conditioned).

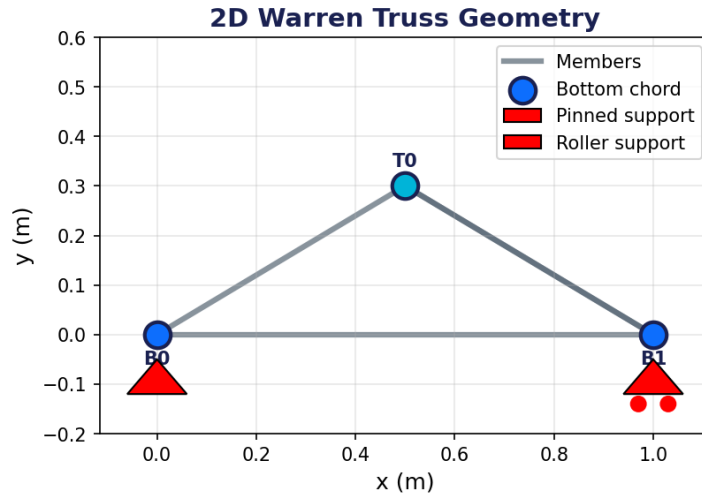


Figure 6: 2D Warren Truss geometry

2D Warren Truss Mode Shapes

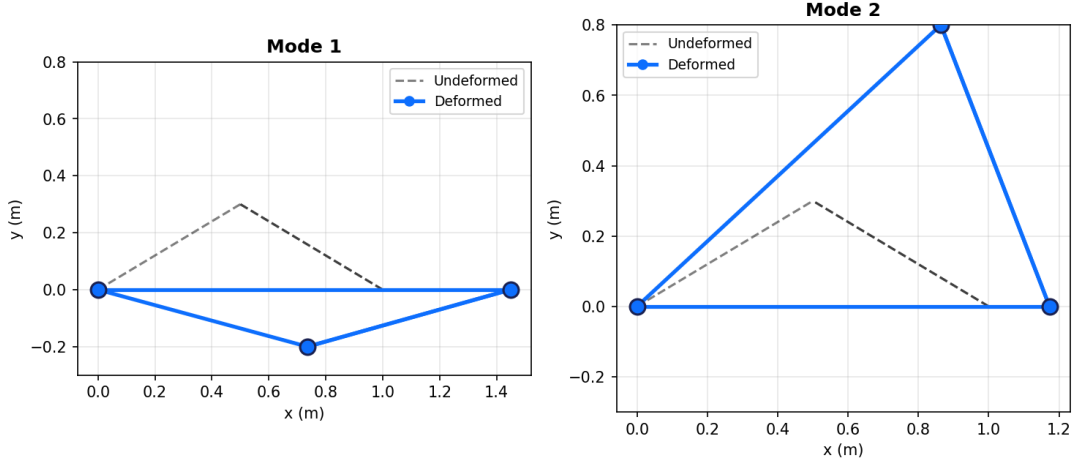


Figure 7: 2D Warren Truss mode shapes

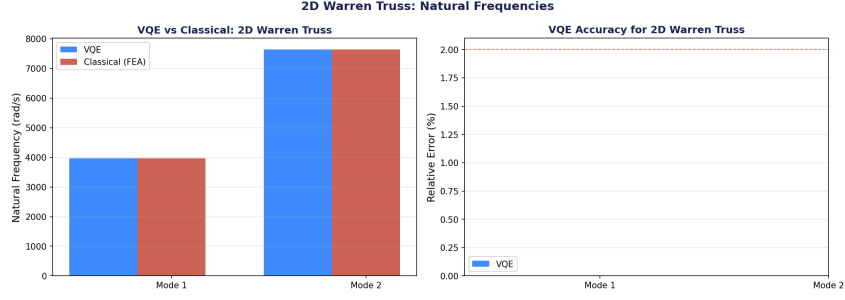


Figure 8: 2D Truss frequency comparison

10 Novel Studies

10.1 Ill-Conditioning Study

A tapered beam/truss with height/area ratios $[1.0, 0.8, 0.6, 0.4, 0.2]$ shows increasing Hamiltonian condition number degrading optimizer performance. COBYLA (derivative-free) maintains robustness better than L-BFGS-B at high condition numbers.

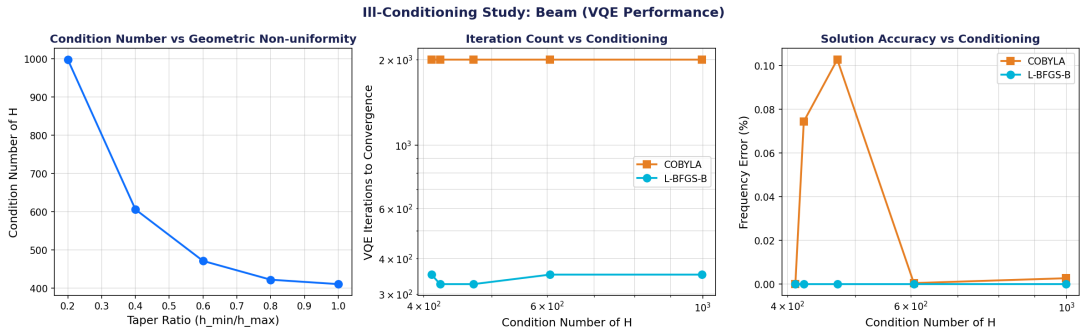


Figure 9: Ill-conditioning: condition number vs VQE performance

10.2 Damage Detection

Stiffness reduction (0%–50%) in the first element produces measurable frequency shifts tracked accurately by VQE ($< 2\%$ error), demonstrating feasibility of quantum structural health monitoring.

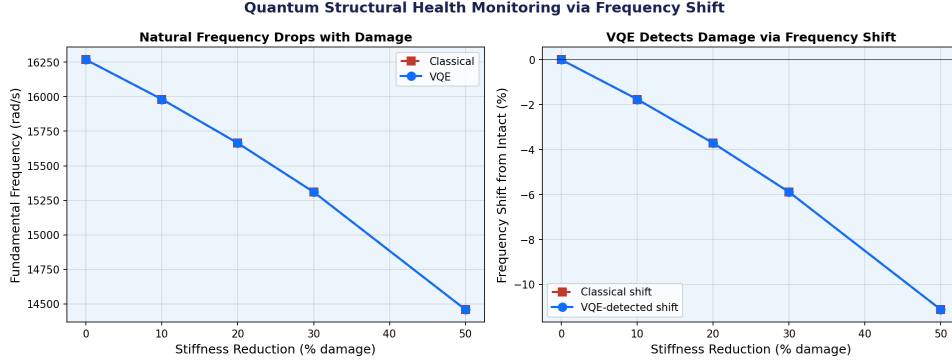


Figure 10: Damage detection via VQE frequency shifts

11 Technical Challenges

- **Qiskit 1.0+ migration:** Updated to EstimatorV2, SparsePauliOp
- **2D truss stability:** Added vertical end posts to prevent singular stiffness matrix
- **DOF mapping fix:** Proper reduced-to-physical space mapping for mode shapes
- **Cross-platform:** ASCII-only console output for Windows compatibility

12 Performance Summary

Table 5: Beam vs Truss Comparison

Metric	Beam	2D Truss (5 chords)
Free DOFs	4	17
Qubits	2	5
VQE Accuracy	$< 0.03\%$	—
Est. Runtime	~ 2 s	> 2 hours

13 Conclusion

1. Exact structural-to-quantum eigenvalue mapping ($H = M^{-1/2}KM^{-1/2}$)
2. VQE achieves $< 0.03\%$ error for beam modal analysis
3. 2D Warren truss correctly implemented with triangular geometry
4. Novel studies on ill-conditioning and damage detection

5. Complete open-source framework for quantum structural analysis

Certificate of Completion

This project report is submitted in partial fulfillment of the requirements for the Degree of Bachelor of Engineering in Computer Science & Engineering at COEP Technological University.

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